Towards Green Acetylene Synthesis and Application

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Abstract

Acetylene, produced from coal or natural gas, was a cornerstone of the chemical industry until the 1960s. However, the long-term global availability of inexpensive petroleum before 1973 accelerated the production of olefins, diminishing acetylene's market prominence due to their similar applications in downstream chemical production. As petroleum prices fluctuate, acetylene has regained economic viability in certain regions, notably accounting for approximately one-third of the global vinyl chloride production, despite its high carbon footprint. Environmental and economic assessment shows that replacing coal with biochar in acetylene-derived vinyl chloride production not only lowers carbon footprint but could also remain economically viable compared to the traditional ethylene route. Despite this potential, research on acetylene has been largely overlooked for decades. This perspective provides the extensive analysis of acetylene production technologies, proposes sustainable production initiatives and quantifies their economic and environmental performance, and explores potential applications. By showcasing this promising trajectory, we seek to rekindle interest and foster collaborative efforts in advancing sustainable acetylene production and broadening its use as a feedstock for bulk chemicals.

Keywords: Acetylene; Catalyst design; Reactor design; Renewable carbon sources; Environmental and economic assessment.

1. Introduction

Acetylene was discovered in 1836 by Edmund Davy through the reaction of water with potassium carbide (**Fig. 1**).[1] Significant study only began after its rediscovery by Marcellin Berthelot who coined the name "acetylene" in 1860 and reported its synthesis using an electric arc in 1862.[2] The first commercial calcium carbide (CaC_2) plant for acetylene (C_2H_2) production was established in 1894.[3] Early C_2H_2 applications were mainly in lighting and oxyacetylene welding due to the brightness and high temperature of its flame.[2] The importance of C_2H_2 in the chemical industry stems from its triple bond, which makes the molecule reactive and offers rich chemical possibilities. Industrial acetaldehyde production via C_2H_2 hydration in 1916, along with studies on C_2H_2 chemistry by Walter Reppe to produce various commodity chemicals, positioned C_2H_2 as a pillar of the chemical industry.[4] Coal-based CaC_2 was the sole source for C_2H_2 production until the development of commercial light hydrocarbon cracking methods, including the electric arc process in 1940, the partial combustion (PC) and thermal pyrolysis (TP) processes in 1950.[5,6]

The birth of the petrochemical industry in the USA and Europe during the 1940s–1950s, coupled with the safer handling of olefins compared to acetylene, facilitated the development of more cost-effective steam cracking technologies for olefin production, gradually displacing the C_2H_2 market.[7,8] Quantitatively, global production of C_2H_2 and ethylene (C_2H_4) reached 3.0 and 3.5 million metric tons per year (Mt y^{-1}) by 1960, respectively,[9–11] and these figures rose to 11 Mt y^{-1} for C_2H_2 and a staggering 225 Mt y^{-1} for C_2H_4 by 2021.[12–14] Between the 1970s and 1980s, PC became the dominant technology for global C_2H_2 production, as other methods were mostly abandoned. In recent years, China has seen a resurgence in CaC_2 production, rising from ca. 3 Mt y^{-1} in 1998 to 28 Mt y^{-1} in 2020.[15] This growth is linked to a significant increase in polyvinyl chloride (PVC) production, from less than 1.6 Mt y^{-1} in 1998 to over 20 Mt y^{-1} in 2020.[16,17] China now leads global C_2H_2 production (>8 Mt y^{-1}), accounting for over 70% of world production.[13,14,18] The historical transition from acetylene to ethylene was driven by the long-term availability of inexpensive oil (**Fig. 1**).[19,20] Nonetheless, as oil prices fluctuate, C_2H_2 production has become economically viable in certain regions where oil prices are high and electricity, coal, or natural gas costs are low.[21] Additionally, the triple bond of C_2H_2 makes it unique for certain applications, such as the production of acetylene black, polyacetylene (Nobel Prize in Chemistry in the year 2000), and Reppe chemicals like 1,4-butanediol, which are difficult to replace with olefins.[22–25]

Acetylene production processes generally show a higher carbon footprint compared to ethylene production.[13,26] As the industry shifts towards sustainability, reducing the carbon footprint of acetylene production through the use of renewable energy, sustainable carbon resources, and new technologies presents an opportunity for its revitalization.[13] However,

research on alternative acetylene production has lagged for decades. By offering a comprehensive analysis of conventional C_2H_2 production methods, summarizing emerging technologies, and outlining pathways toward sustainable processes, this perspective aims to rekindle interest and foster collaborative efforts in advancing green acetylene production while addressing the associated challenges and opportunities.

2. Production of acetylene

2.1. Summary of acetylene production routes and its applications

Acetylene can be produced from fossil feedstocks such as coal, natural gas, and oil using four primary technologies, namely CaC_2, partial combustion (PC), plasma, and thermal pyrolysis (TP) (**Fig. 2**).[27] Among these, oil-based routes are yet to reach the commercial stage. Plasma technologies were largely replaced by PC processes utilizing natural gas as feedstock due to their extremely high energy consumption.[5,28] TP of hydrocarbons was commercialized via the Wulff process;[6] however, these plants eventually shut down due to a slow heating rate and significant soot formation compared to PC. Notably, C_2H_2 production is region-dependent, with China producing over 8 Mt y^{-1} primarily from the CaC_2 route (~90%) due to its large coal reserves, while other countries produce less than 3 Mt y^{-1}, mainly via PC.[9,13]

The major application of C_2H_2 is as a feedstock in the chemical industry,[29] with key downstream products summarized in **Fig. 2**. Among all, vinyl chloride has the highest production volume, reaching approximately 50 Mt y^{-1},[30] predominantly used to produce PVC with a market size of USD 68 billion. Globally, the PVC market is dominated by the C_2H_4-based route, accounting for about 65% of PVC output.[31] However, in China, ca. 78% of PVC production relies on the C_2H_2 route, with the remainder produced via the C_2H_4 route.[21] The market for acetylene black, valued at approximately USD 0.3 billion, is expected to grow with the global shift towards electric vehicles due to its effectiveness as a conductive additive for electrodes.[9,32]

Another critical application of C_2H_2 is the production of 1,4-butanediol via the classical Reppe process, which involves the reaction of acetylene with formaldehyde. Although several complex and multi-step alternative routes exist, with maleic anhydride or 1,4-diacetoxy-2-butene as the intermediate, the recent resurgence of the C_2H_2 route has significantly lowered market prices. This route now accounts for 40% of global production (2–2.5 Mt y^{-1}), resulting in overcapacity of 1,4-butanediol.[33,34] This highlights the distinct advantages of C_2H_2 as a feedstock, which typically enables more streamlined downstream synthesis due to its reactive C≡C triple bond. While the global industrial production of vinyl acetate, acetaldehyde, acetic acid, acrylic acid, and acrylonitrile has largely shifted to routes utilizing C_2H_4, propylene, and methanol (**Fig. 2**), vinyl acetate production in China primarily relies on the C_2H_2 route.[35-41]

2.2. Commercial acetylene production routes

The CaC_2 route involves two reaction steps, as shown below. Unlike other methods, C_2H_2 is produced at room temperature without thermodynamic limitations. However, the production of CaC_2 is highly endothermic through the reaction of carbon and lime.[42]

$$3C + CaO \rightarrow CaC_2 + CO \qquad \Delta H = 465\ \text{kJ mol}^{-1}$$

$$CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2 \qquad \Delta H = -127\ \text{kJ mol}^{-1}$$

Despite its high selectivity for C_2H_2 and the low equipment investment, several disadvantages exist (**Fig. 3a**): (i) the CaC_2 formation reaction requires temperatures of 2000–2300 °C to overcome the high endothermicity, leading to the electricity consumption of ca. 3–3.5 kWh kg^{-1} CaC_2 and approximately 10.6 kWh kg^{-1} C_2H_2;[43] (ii) large amounts of coal is consumed, with one-third of carbon resources converted to CO;[13] (iii) significant CO_2 emissions of 3.6 kg result from the decomposition of limestone ($CaCO_3$) and the combustion of byproduct;[13] (iv) large amounts of solid slag waste, mostly $Ca(OH)_2$, are produced.[13] Due to these disadvantages and the transition from C_2H_2 to olefins, most CaC_2 plants in Western countries have shut down. However, in China, where coal reserves are abundant and imported oil for olefin production is costly, the CaC_2 route remains significant, reflected by significant increase in CaC_2 production from 3 to 28 Mt y^{-1} in the 21st century accounting for more than 96% of the global production.[15,44]

Direct conversion of CH_4 to C_2 hydrocarbons is highly endothermic, with the CH_4 to C_2H_2 reaction being the most endothermic, having a standard enthalpy of 376 kJ mol^{-1}.[27] This corresponds to the theoretical limit for the energy requirement of 4.0 kWh kg^{-1} C_2H_2.[45]

$$2CH_4 \rightarrow C_2H_2 + 3H_2 \qquad \Delta H = 376\ \text{kJ mol}^{-1}$$

$$2CH_4 \rightarrow C_2H_4 + 2H_2 \qquad \Delta H = 202\ \text{kJ mol}^{-1}$$

$$2CH_4 \rightarrow C_2H_6 + H_2 \qquad \Delta H = 65\ \text{kJ mol}^{-1}$$

Thermodynamic data show that CH_4 is the most stable hydrocarbon at low temperatures, while C_2H_2 is the least stable (**Fig. 3b**).[27] With increasing temperature, the free energy of C_2H_2 decreases, making its formation more favorable than other hydrocarbons above 1230 °C.[27,46] However, high temperatures also promote C_2H_2 decomposition to carbon and hydrogen. Thus, achieving reasonable yields from CH_4 requires high reaction temperatures and rapid quenching of the C_2H_2 stream, such as in commercial PC and thermal plasma technologies. For PC, approximately 60% of CH_4 combusts with O_2 in the burner to stabilize the flame at a temperature above 1500 °C, facilitating the conversion of the remaining CH_4.[27] The reaction gas exits the reactor within milliseconds, followed by rapid quenching with water or oil sprays. Thermal plasma, known as the electric arc,[47] achieves temperatures of up to 20,000 °C at the arc center, while the electrode walls remain around 600 °C.[5] **Fig. 3c** compares these two technologies.[5,9,13,27] The thermal plasma method yields a higher C_2H_2 concentration in the cracked gas, lower natural gas consumption, and reduced direct CO_2 emissions, but requires

significantly higher electricity (12.1 kWh kg^{-1} C_2H_2) and leads to extensive soot formation.[5,48] With the increasing availability of green electricity, thermal plasma has garnered more attention.[13,21,49]

2.3. Emerging C_2H_2 synthesis routes via the plasma process

Coal-to-C_2H_2 production via thermal plasma technologies has seen significant development in China recently, driven by high carbon footprint associated with the CaC_2 route. Since 2014, Zhejiang University and Xinjiang Yuehetai Chemical Technology Co. Ltd. have collaborated to develop 2 MW and 10 MW H_2 plasma systems for C_2H_2 production, claiming that the costs are over 20% lower than conventional CaC_2 routes, with a significantly reduced carbon footprint.[50,51] In contrast to thermal plasma, non-thermal plasmas preferentially channel energy to electrons, creating a much higher electron temperature than that of ions and neutrals. When the neutral temperature is close to room temperature, the plasma is termed “cold plasma” (e.g., dielectric barrier discharge (DBD), corona, and pulsed plasma). In contrast, microwave and spark plasmas can heat neutrals to several thousand degrees Celsius, making thermal activation significant. These are known as “warm plasmas”.[52] The main advantage of cold plasma is that it eliminates the need for quenching, as C_2H_2 is produced at low temperatures. In contrast, warm plasma typically achieves higher CH_4 conversion due to the higher neutral temperatures (**Supplementary Table 1**). Feedstocks typically include pure CH_4 or CH_4 diluted in O_2,[53,54] H_2,[55] and CO_2.[56]

Reactor designs for non-thermal plasma technologies vary widely (**Fig. 4a**), necessitating case-by-case considerations to optimize energy efficiency. Generally, electrode-based plasmas require electrodes for gas ionization to form plasma conditions, while microwave plasma induces ionization via microwave irradiation, thus avoiding soot deposition on electrodes, which can affect plasma stability.[5] DBD uses a dielectric barrier to prevent arcing and ensure uniform plasma discharge, while corona reactors rely on a pointed electrode to generate a localized, non-uniform electric field for plasma formation. In addition to reactor design, catalysts also play an important role in non-thermal plasma, with DBD and microwave being the most commonly used (**Supplementary Table 1**). For instance, in DBD plasma, adding the MFM-300(Fe) catalyst increased C_2H_2 selectivity from 17% to 38% at a CH_4 conversion of <3%.[57] In contrast, in microwave plasma, microwave irradiation seems more essential for C_2H_2 production than catalysts in most cases.[58,59] While extensive literature emphasizes the significance of non-thermal plasma in C_2H_2 production, detailed energy requirements are sometimes not provided, an essential indicator for evaluating relevant technologies. A possible reason is that it requires substantial energy input, similar to or even exceeding thermal plasma processes (**Fig. 4d**).

2.4. Advancements in catalyst design for the thermal pyrolysis process

For thermal pyrolysis (TP), the oxidative coupling of methane (OCM) and non-oxidative coupling of methane (NCM) can produce C_2 species. However, the slower heat transfer in TP, compared to PC limits its commercial viability.[27] Therefore, TP could only be competitive if reaction temperatures are reduced using catalysts. OCM is thermodynamically more favorable than NCM, as it requires lower reaction temperatures (450–1000 °C), but faces challenges in preventing C_2H_2 conversion or overoxidation to CO_x in the presence of O_2.[60] Additionally, at low temperatures, the reaction of CH_4 to C_2H_2 becomes less thermodynamically favorable compared to the formation of other hydrocarbons (**Fig. 3b**). As a result, C_2H_2 production from catalytic TP primarily comes from NCM rather than OCM (**Supplementary Table 2**). Advanced reactor designs, including membrane (**Fig. 4b**) or millisecond reactors, have been explored for NCM to enhance CH_4 conversion by shifting the reaction equilibrium toward the product side or reducing soot formation due to shortened contact time.[61,62]

The highest C_2H_2 selectivity reported is 35% with approximately 14% CH_4 conversion at 975 °C on a Pt_1/CeO_2 (Pt_1: Pt single atom) catalyst,[63] outperforming pristine CeO_2, Pt_n/CeO_2 (Pt_n: Pt nanoparticles). However, recent studies have revealed that Pt_1 in Pt_1/CeO_2 tends to sinter under reaction conditions, and the obtained acetylene selectivity is relatively low.[64–66] A CH_4 conversion of 63% is achieved with Pt on α-Al_2O_3 monolith (**Fig. 4c**), but a significant temperature gradient in the monolith from 200 to 1500 °C is observed, which is already close to the temperature in PC.[67] Metal carbides and Fe_1 catalysts (**Fig. 4e**) also show good promise,[68,69] though CH_4 conversion is typically below 30% at temperatures under 1100 °C. Additionally, coke deposition and soot formation are significant in catalytic TP, although millisecond reactor designs can partially address this issue.[62] To date, achieving both high CH_4 conversion and high C_2H_2 selectivity simultaneously via catalytic TP remains challenging, with performance significantly worse than plasma processes (**Fig. 4c, Supplementary Table 1 and 2**). The presence of byproducts, such as other C_2, C_3-C_5 hydrocarbons, aromatics, and coke/soot, limits C_2H_2 formation.[69,70] Therefore, converting CH_4 into a broader range of products, such as C_2 or even C_{2+} hydrocarbons, may be a more practical route for future CH_4 utilization via catalytic TP.

3. The drive for acetylene's revival

The primary application of acetylene is in the production of vinyl chloride monomer (VCM) and subsequently PVC, a process that accounts for approximately 35% of the global PVC production (**Fig. 2**). A production cost analysis (**Fig. 5a**), based on current data (refer to the **Supplementary Notes** for a detailed methodology), shows that the coal-based acetylene to VCM (ATV) scenario has a total cost of 0.6 USD kg^{-1} of VCM, making it more economical than the fossil-based (naphtha) ethylene balanced process (EBP) showing a total cost of

1.1 USD kg^{-1}. However, the climate change impacts (**Fig. 5b**) associated with ATV are notably higher than those of the EBP route (5.7 vs. 2.4 kg CO_2-eq kg^{-1}, assuming the same feedstock alternatives, respectively), primarily due to the more energy-intensive production of C_2H_2. This economic and environmental trade-off has led to the coexistence of both ATV and EBP routes in the market. **Supplementary Tables 3–5** provide the mass and energy balance, life cycle inventories, and feedstock and utility prices, respectively, which were used to estimate the production costs and climate change impacts of VCM.

As the chemical industry moves towards sustainability, both ATV and EBP routes must explore more environmentally friendly alternatives, particularly for the ATV route. By utilizing biochar-based C_2H_2 and C_2H_4 derived from the dehydration of bioethanol, a green scenario could emerge. The latter is already being deployed in countries like Brazil, using sugarcane as a sustainable feedstock. Interestingly, the cost of the biochar-based ATV route only slightly increases to 0.8 USD kg^{-1} when considering poplar as biomass source, while the cost of bioethanol-based EBP process rises significantly to 2.3 USD kg^{-1} (**Fig. 5a**), considering corn stover as feedstock. Although the climate change impacts in the green scenario remain higher for the ATV compared to the EBP route (1.7 vs. 0.8 kg CO_2-eq kg^{-1}), both routes show a significant reduction in carbon footprint simply by replacing the feedstocks. Notably, compared to the fossil-based EBP route, the biochar-based ATV route exhibits lower costs and reduced climate change impacts. Therefore, developing green acetylene production methods could drive a transition in PVC production from EBP to ATV routes, potentially revitalizing the acetylene industry. Furthermore, with PVC demand projected to grow at an annual rate of 3.8% until 2031 (data from Statista), the sustainable ATV route could gain momentum, particularly if the conventional EBP route does not achieve significant breakthroughs. Acetylene's market share could also expand through increased utilization in producing 1,4-butanediol, acetylene black, polyacetylene, and vinyl acetate, as well as by reclaiming its role in manufacturing products like acetaldehyde, acetic acid, acrylic acid, and acrylonitrile (**Fig. 2**). Recent advancements in acetylene-based technologies further highlight its industrial relevance, such as the electrocatalytic conversion of C_2H_2 to butadiene or C_2H_4,[71,72,73] the photocatalytic hydrochlorination of C_2H_2 to VCM,[74] and the coupling of C_2H_2 with CO to synthesize esters and acids.[75,76]

4. Renewable carbon resources for acetylene synthesis

Given the performance of biochar-based C_2H_2 in the ATV route (**Fig. 5a, 5b**), it is essential to explore sustainable C_2H_2 production methods using renewable resources (e.g., biochar, biomethane, CO_2, plastic) and innovative production technologies to replace traditional fossil-based approaches (**Fig. 5c**). Additionally, integrating renewable electricity (e.g., wind and solar) into these processes is crucial for further advancing sustainability. **Fig. 5d** illustrates

potential combinations of renewable carbon resources and relevant technologies for C_2H_2 synthesis. Biochar, derived from biomass pyrolysis, has been proposed for C_2H_2 synthesis via the CaC_2 route with a lower carbon footprint. However, it typically exhibits lower carbon and energy efficiency compared to the coal-based CaC_2 route due to the lower energy density of biomass and the generation of volatile components during pyrolysis.[14,77] Direct TP of various biomass feedstocks to C_2H_2 has been widely studied but is limited by the selectivity towards C_2H_2.[47] Biomass as a substitute for coal in the plasma route has also been explored, such as lignin and hemicellulose cracking in a hydrogen plasma, further reducing the carbon footprint.[78]

The production of C_2H_2 from polyolefins and polyethylene terephthalate (PET) via rotating arc plasma has proven feasible, achieving full carbon conversion and C_2H_2 selectivity of 55% for PET and 80% for polyolefins.[79,80] However, the specific energy requirement is approximately 20 kWh kg^{-1} C_2H_2. Biogas, a renewable carbon feedstock composed mainly of CH_4 and CO_2, could replace natural gas for C_2H_2 synthesis with a lower carbon footprint. For instance, the conversion of biogas-like mixtures of CH_4 and CO_2 into C_2 products can be achieved using DBD cold plasma, with C_2H_6 as the dominant product.[81] Moreover, the addition of a catalyst (Ni–K/Al_2O_3) and an increase in the reaction temperature above 220 °C can enhance C_2H_2 selectivity from biogas.[82] The abundant availability of CH_4, or its presence as a byproduct of CO_2 utilization in biogas, presents an opportunity to utilize biomethane for C_2H_2 production (**Fig. 5d**).[29]

CO_2 in biogas or directly from the air can be integrated into the C_2H_2 production process via CO_2-CO or CO_2-C_2H_2 cycles. A sustainable route for C_2H_2 production, coupled with CO_2 capture and conversion to CO (CO_2-CO cycle), is based on the $BaCO_3$-BaC_2-$Ba(OH)_2$-$BaCO_3$ cycle (**Fig. 5e**).[83] Unlike the CaC_2 route, where CaO is obtained from $CaCO_3$ decomposition, leading to CO_2 emissions, the CO_2-CO cycle does not directly emit CO_2. Additionally, BaC_2 synthesis requires lower temperatures, and less energy input compared to CaC_2. Notably, the carbon from CO_2 and half of the carbon from coal or biochar converts to CO instead of C_2H_2 in the CO_2-CO cycle (**Fig. 5e**), lowering atom economy. In contrast, the proposed CO_2-C_2H_2 concept cycle directly transforms CO_2 into C_2H_2 via a cyclic electrochemical-assisted strategy (**Fig. 5f**).[84] This CO_2-mediated cycle involves three steps that hold the most significant promise for sustainability. However, it is still in the early stages, as the recycling of Li_2O or $LiCO_3$ is yet to be addressed.

In the context of renewable carbon feedstock, projections suggest that by 2050, the annual availability of biochar and biomethane will reach 0.3–0.5 Gt and 0.1 Gt, respectively.[85,86] This implies that only 4–7% of the available biochar (via the CaC_2 process) or 26% of the biomethane (via the PC route, and significantly less via plasma routes) would

be sufficient to meet current global C_2H_2 demands, estimated at about 11 Mt y^{-1}. Therefore, incorporating biochar or biomethane as potential alternatives in green acetylene production holds significant promise.

5. Environmental and economic assessment of acetylene synthesis routes

In light of the framework proposed to guide the development of sustainable chemical processes by prioritizing the environmental, economic, and equity (EEE) pillars,[87] this section evaluates the environmental impact and economic viability of various acetylene synthesis technologies through techno-economic and life cycle assessment (LCA) analyses. We compare these technologies using both fossil and renewable carbon resources, assessing their environmental impacts within a future decarbonized scenario. For detailed insights into our methodology, along with the safety risks linked to each acetylene production route, please refer to the **Supplementary Notes**.

We specifically compare the climate change impacts shown in **Fig. 6a** and production costs (excluding capital expenditure) in **Fig. 6b**, both associated with the production of one kilogram of C_2H_2 using three different processes: CaC_2, PC, and plasma (both thermal and non-thermal). **Supplementary Tables 6–8** present the mass and energy balance, life cycle inventories, and feedstock and utility prices, respectively, which were utilized to estimate the production costs and climate change impacts of C_2H_2. We evaluated two scenarios for all production technologies: fossil-based and green. For the CaC_2 process, we used coke (fossil) and biochar (green) as feedstock. For the technologies utilizing methane as a feedstock in the fossil scenario (PC and plasma routes), we replaced methane with biomethane in the green scenario. Furthermore, we conducted a prospective LCA to evaluate the projected climate change impacts for the year 2050 under a decarbonized scenario.

The highest climate change impacts are observed in the CaC_2 production scenario (12.9 kg CO_2-eq kg^{-1}) due to its heavy reliance on coke and electricity to produce calcium carbide, as well as substantial direct emissions. In contrast, the PC route has the lowest emissions (7.8 kg CO_2-eq kg^{-1}), utilizing methane as the primary feedstock and resulting in much lower direct emissions compared to the CaC_2 scenario. Both plasma technologies exhibit higher climate change impacts than the PC route, primarily due to the high electricity requirements of these pathways. However, emissions are significantly reduced across all green scenarios. Notably, the PC route displays a negative climate change impact of –6.1 kg CO_2-eq kg^{-1} due to the utilization of biomethane feedstock, which results in negative net contributions to the overall impacts (on a cradle-to-gate basis) due to the biogenic nature of the carbon.[88] Similarly, the impacts for the green CaC_2 route show a 75% reduction due to the shift from coke to carbon-negative biochar.[89] In contrast to the significant reductions observed in the CaC_2 and PC routes, both plasma technologies show only a 45% decrease

under the green scenario, due to their heavy reliance on grid electricity, which is yet to be decarbonized.

In a future decarbonized scenario projected for 2050, emissions are expected to decrease significantly, primarily due to the adoption of a decarbonized electricity mix.[90] The most optimistic scenario (1.5 °C – RCP 1.9) is highlighted in **Fig. 6a** (purple scatter).[91] The CaC_2 route demonstrates a 63% reduction in emissions compared to current impacts under the fossil scenario, and a 283% reduction under the green scenario. Similar reductions are observed across all production routes. These results underscore the potential of a decarbonized grid to mitigate climate change impacts, even in fossil-based production methods, and especially in the green scenario, as biochar also uses decarbonized electricity compared to the current scenario.

Regarding production costs within the fossil scenarios, the CaC_2 route has the lowest cost (1.2 USD kg^{-1}), primarily driven by electricity. The PC route is more expensive (2.0 USD kg^{-1}), largely due to methane costs. Similarly, both plasma routes have comparable costs, averaging 1.7 USD kg^{-1}, predominantly due to electricity. In the green scenario, the CaC_2 route remains the most economical, with a 32% increase in costs due to the slightly higher cost of biochar compared to coke. The green PC scenario is the most expensive route, with a 340% increase in costs, driven by the substantial cost of biomethane (five times more expensive than methane). Similarly, plasma routes show at least a 110% increase in costs.

In the short term, the focus should be on optimizing the existing CaC_2 production process by transitioning from coke to a blend of coke and biochar, ensuring stable biochar supply chains to maintain cost-effectiveness. The lower costs of biochar and its significant reduction in climate change impacts make the green CaC_2 route the most favorable option for producing C_2H_2. In the long term, transitioning to renewable sources and engaging in grid decarbonization initiatives for electricity will further reduce the climate change impacts of plasma-based C_2H_2 production at lower costs. Therefore, comprehensive analysis is essential for engaging with stakeholders and advocating for policies that steer the chemical industry toward sustainable and economically viable C_2H_2 production.

6. Outlook

Transitioning to a sustainable chemical industry is a global imperative that presents an opportunity to revitalize the acetylene industry. To achieve this, focus on both C_2H_2 production and its chemistry is essential. For production, two main strategies are considered: integrating mature technologies with renewable carbon sources and decarbonized electricity to reduce the carbon footprint, and exploring emerging technologies to improve C_2H_2 yield and energy efficiency. Replacing natural gas with more expensive biomethane is not suitable for the PC route, as it requires nearly three times more biomethane than plasma technologies.

Consequently, biochar-based CaC_2 and biomethane-based plasma routes are expected to become more prominent. Their development will depend on the availability and cost of biochar and biomethane, with competitive uses emerging (e.g., biochar for carbon removal and biomethane for defossilizing industrial heating). Given the explosive potential of C_2H_2, on-site generation for downstream applications is advantageous.[27] The CaC_2 route offers a significant advantage by eliminating the risk of explosion through the transport of solid CaC_2 instead of C_2H_2, positioning the biochar-based CaC_2 route favorably for the future.

Emerging technologies, such as non-thermal plasma, face commercialization challenges, requiring advancements in reactor design, catalyst development, and a deeper understanding of reaction mechanisms. It demonstrates performance close to commercial thermal plasma but typically requires similar or even higher energy input and more complex equipment. While catalysts can favor C_2H_2 production, their use does not appear to reduce energy requirements, and the design of catalytic plasma reactors remains complex and insufficiently explored. Therefore, future research should aim to improve C_2H_2 yield, energy efficiency, and equipment lifespan.

Unlike non-thermal plasma, C_2H_2 yield through TP remains below 10% (**Fig. 4c**) due to thermodynamic limitations at low temperatures. To improve activity, selectivity, and catalyst stability, advanced catalyst designs (e.g., single-atom Pt_1 or Fe_1) and reactor designs (e.g., membrane or millisecond reactors) have been developed. Despite decades of effort, catalytic TP is not yet commercially viable and is therefore more practical for selective conversion to C_2 products or mixtures of C_2 and other hydrocarbons. Additionally, kinetic studies and advanced *operando* characterization will be valuable in elucidating the complex CH_4 reaction mechanisms, likely involving coupled surface and gas-phase chemistry. Designing high-temperature reactors that operate up to 1100 °C for synchrotron-based spectroscopy studies will present a significant challenge in the future. Plasma-assisted plastic upgrading to C_2H_2 is promising but currently requires significant energy input (20 kWh kg^{-1} C_2H_2), making it impractical without further breakthrough.[79,80] Similarly, transforming CO_2 and H_2O into C_2H_2 and O_2 via the CO_2-C_2H_2 concept cycle is an ideal but nascent approach.[84]

Revitalizing the C_2H_2 industry also requires a substantial market for C_2H_2 applications. The development of sustainable C_2H_2 production could prompt a shift in VCM production from ethylene to acetylene routes, driven by environmental and economic assessments. Moreover, the longstanding challenge in C_2H_2-derived VCM production–specifically the use of mercury catalysts–has been addressed due to the recent commercialization of an Au/C catalyst.[92,93] The anticipated growth in the PVC market, along with the potential shift from ethylene- to acetylene-based production, exemplifies the possible resurgence of the acetylene industry. Since C_2H_2 chemistry was only extensively studied by Reppe in the mid-20th century, there

remains significant untapped potential beyond VCM production and other conventional applications. Emerging technologies, such as electrocatalysis and photocatalysis, along with novel acetylene chemistry (e.g., its coupling with CO or CO_2), warrant greater attention. Overall, we believe the expansion of the downstream market, combined with advancements in green acetylene production, could serve as a catalyst for the revitalization of the C_2H_2 industry.

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Author contributions

Z.Z. and J.P.-R. conceived the perspective. Z.Z. and A.N. produced figures. A.N. and G.G.G. completed the life cycle assessment and techno-economic study. Z.Z. and A.N. co-wrote the manuscript. G.G.G. and J.P.-R. revised the manuscript. J.P.-R. supervised the project. All authors reviewed, edited and enriched each section of the perspective.

Competing interests

The authors declare no competing interests.

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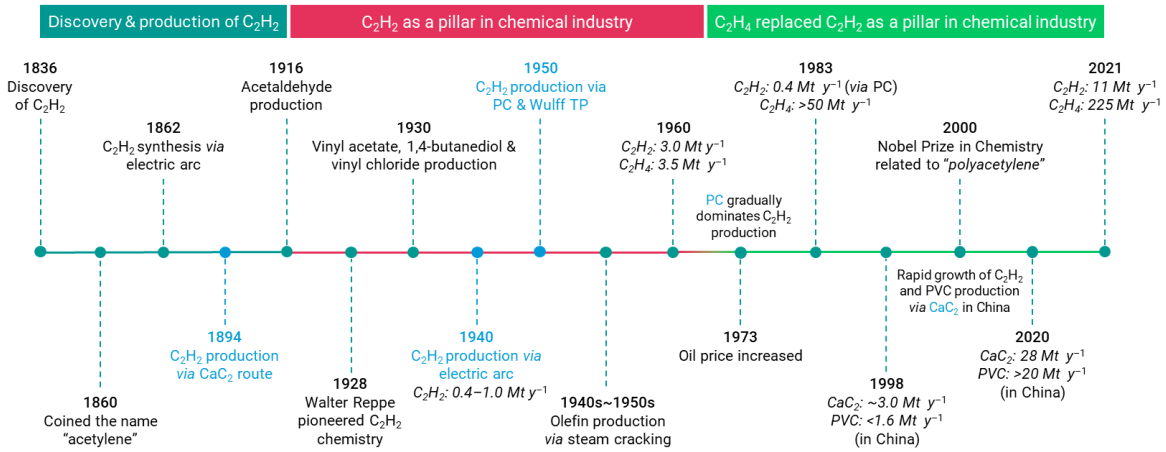


Fig. 1 | The acetylene timeline. The timeline illustrates the evolution of various industrial production technologies for C_2H_2 since its discovery, advancements in C_2H_2 chemistry, global production changes of relevant chemicals at key historical points, and the gradual market transition from C_2H_2 to C_2H_4.

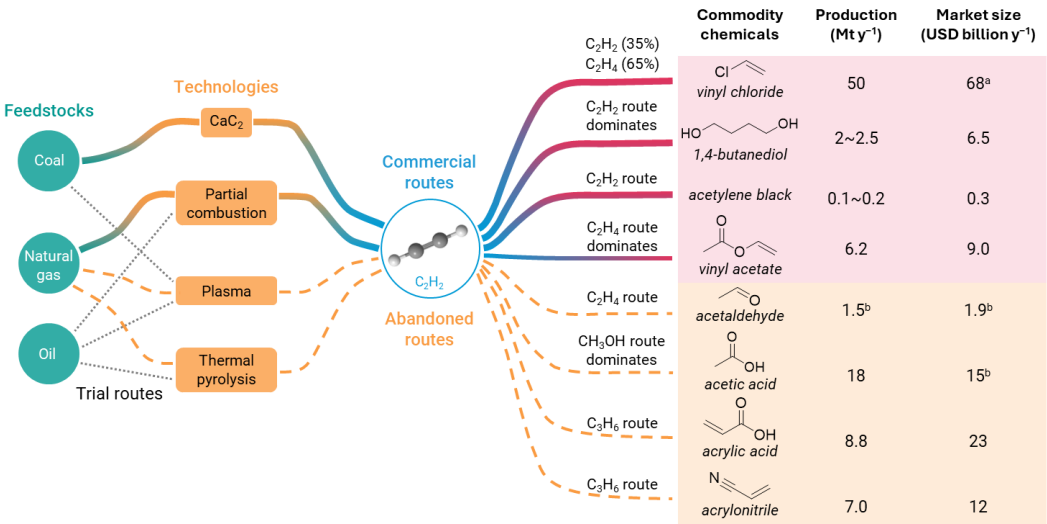
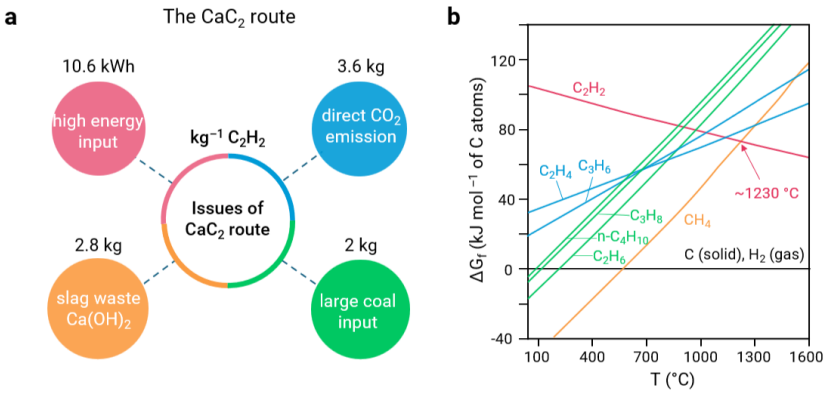


Fig. 2 | C_2H_2 production routes, past and current applications. The figure introduces various C_2H_2 production routes using fossil-derived feedstocks, including coal, natural gas, and oil, along with several technologies such as CaC_2, PC, plasma, and TP. The figure highlights major industrial applications of C_2H_2 and provides data on approximate global annual production (Mt y^{-1}) and market size (USD billion y^{-1}). It distinguishes between commercial routes (currently in use), abandoned routes (past commercial processes replaced by other methods), and trial routes (processes with research interest but never largely commercialized). [a]Since almost all VCM demand is for PVC production, this number reflects the global market size of PVC. [a,b]Data from Statista.



c Comparison of partial combustion and thermal plasma routes

C_2H_2 in cracked gas	8 vol%	11~15 vol%
energy input (kg^{-1} C_2H_2)	3.3 kWh	12.1 kWh
natural gas input (kg^{-1} C_2H_2)	5.7 Nm^3	2.0 Nm^3
CO_2 emission (kg^{-1} C_2H_2)	0.8 kg	~
soot formation (kg^{-1} C_2H_2)	50 mg	450 mg

Fig. 3 | Commercial acetylene production routes. a. Four major issues associated with the CaC_2 route, including high energy input, high direct CO_2 emissions, production of slag waste, and large coal input. **b.** Gibbs free energy of formation per carbon atom for several hydrocarbons from the elements as a function of temperature. **c.** Comparison of partial combustion and thermal plasma routes. Panel **b.** adapted with permission from ref. 32, Wiley.

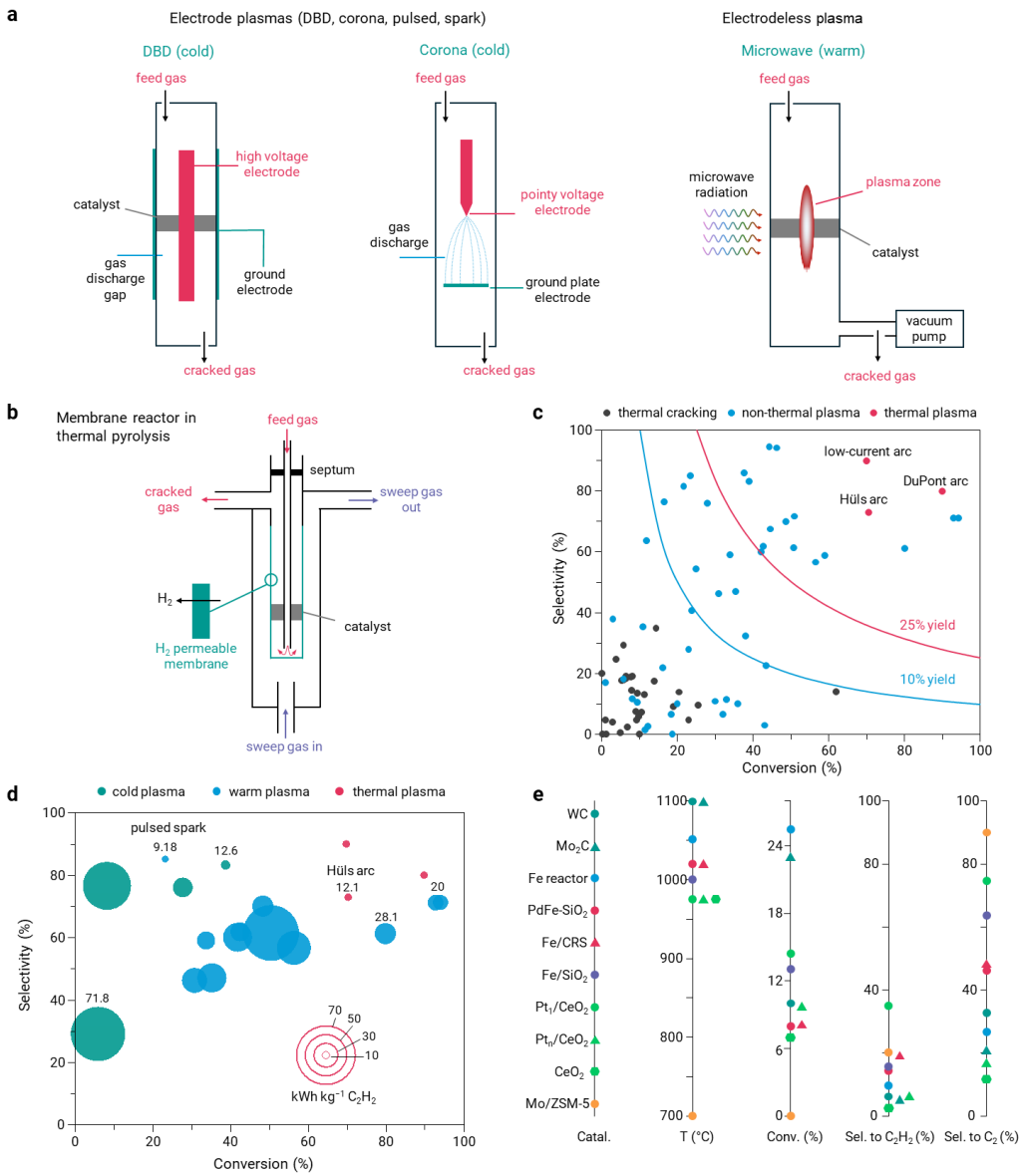


Fig. 4 | Emerging acetylene synthesis routes. **a.** Electrode- and electrodeless-based non-thermal plasma reactors.[5,69] **b.** Structure of membrane reactor in TP.[65] **c.** Conversion-selectivity trend in TP, non-thermal and thermal plasma. **d.** Conversion-selectivity trend and energy requirement (kWh kg^{-1} C_2H_2) for representative non-thermal (cold and warm) and thermal plasma routes. **e.** Representative catalyst type, reaction temperature, CH_4 conversion, and C_2H_2 and C_2 selectivity in non-oxidative CH_4 conversion. Panels adapted with permission from: **a.** ref. 5, Royal Society of Chemistry; **b.** ref. 65, Wiley.

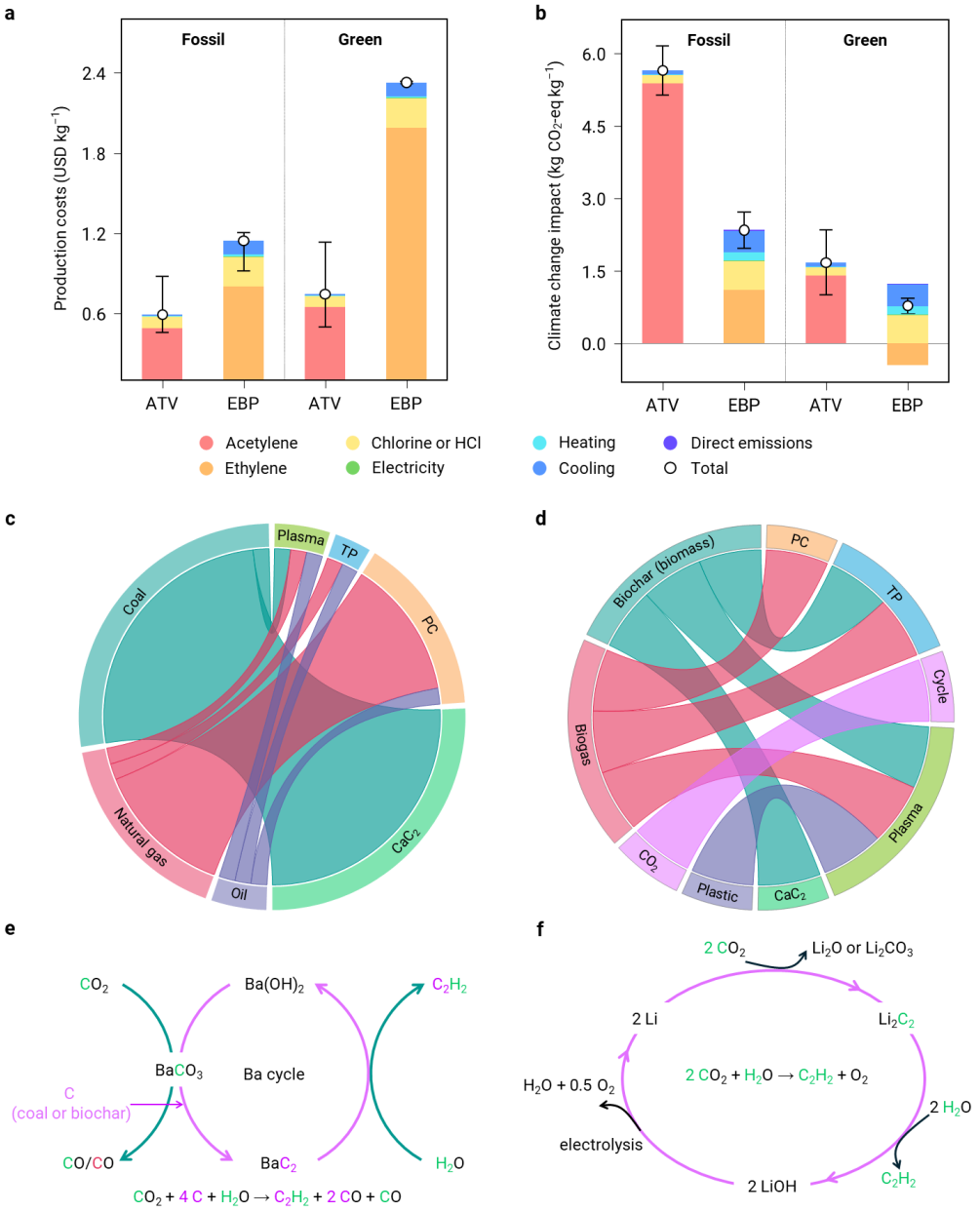


Fig. 5 | Potential green C_2H_2 synthesis routes. a. Production costs (only operational costs) and **b.** climate change impacts per kg of VCM *via* the acetylene-to-VCM (ATV) route and the ethylene-based balanced process (EBP). Results for both the fossil and green routes, depending on the feedstock source. Uncertainties (error bars) in both climate change impacts and production costs are displayed (see **Supplementary Notes** for details). **c.** Coal, natural gas, and oil-based technologies, including CaC_2, PC, TP, and plasma routes. The commercial routes, coal-based CaC_2 and natural gas-based PC, have a larger width due to their extensive use. **d.** biochar, biogas, CO_2, and plastic-based technologies, including CaC_2, plasma, cycle, TP, and PC. Renewable source-based routes are not yet commercialized, so their width remains uniform. CO_2-based concept cycles include **e.** BaC_2-based route coupling CO_2 capture and CO production (CO_2-CO cycle), and **f.** Li_2C_2-based electrochemical-assisted three-step cycle from LiOH to Li, with Li reacting with CO_2 to form Li_2C_2, followed by reaction

with H_2O to produce C_2H_2 (CO_2-C_2H_2 cycle). Panels adapted with permission from: **e.** ref. 49, Royal Society of Chemistry; **f.** ref. 57, Royal Society of Chemistry.

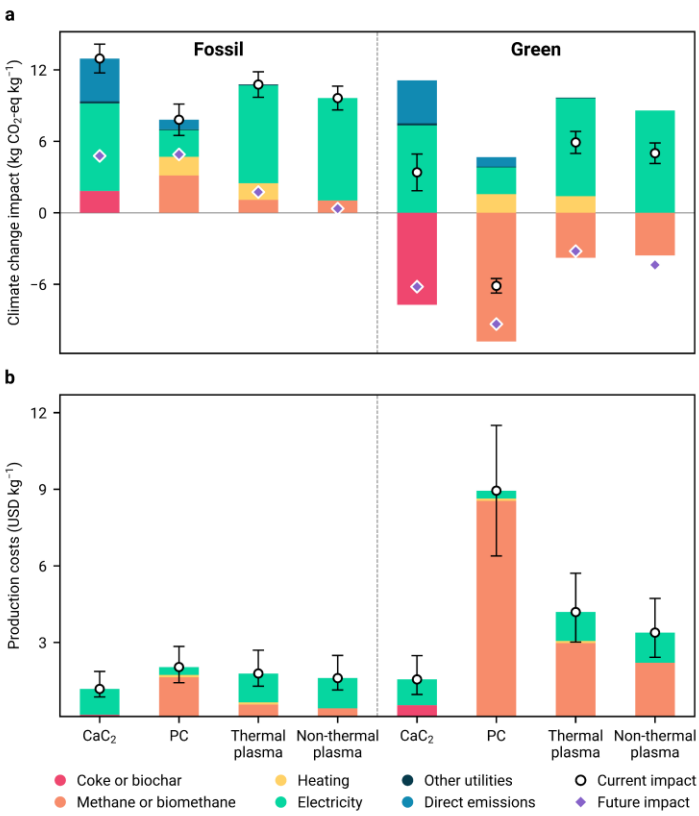


Fig. 6 | Environmental and economic comparison of C_2H_2 manufacture. a. Climate change impacts and **b.** production costs (only operational costs) per kg of C_2H_2 via the CaC_2 route and the PC or plasma routes. Biochar (from poplar), as an alternative to coke for the CaC_2 route, and biomethane, as an alternative to methane for the PC and plasma technologies, are evaluated for the green routes. Future projections (for 2050) of climate change impacts for all technologies are presented to illustrate the potential reduction in impacts under a decarbonized electricity mix scenario. Uncertainties (error bars) in both climate change impacts and production costs are displayed (see **Supplementary Notes** for details).